Self-Induction – behavior of a Coil

You saw in chapter 8 (Electromagnetic Induction) that when the flux ϕ of the magnetic field that crosses a coil varies, the coil reaction is then to generate an electromotive force (voltage) to oppose or resist the flux change. The induced e.m.f can be calculated by Faraday's Law:

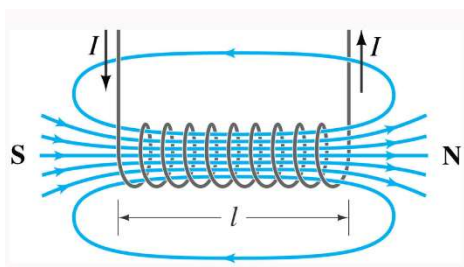
$$e = -\frac{d\phi}{dt}$$

In addition, in grade 11, you learned that a coil traversed by an electric current i becomes an electromagnet: the coil produces its own magnetic field. The magnitude of this field in the space inside the coil is given by:

$$B = \frac{\mu_0 N i}{l}$$

Where

- μ_0 is the permeability of vacuum inside the coil (similar to ε_0 the permittivity of vacuum inside a capacitor). $\mu_0 = 4\pi \times 10^{-7}\ (SI)$.
- N is the number of turns for the coil.
- l is the length of the coil. $l(m)$.
- i is the magnitude of the current flowing through the coil. $i(A); B(T)$.



The flux of the magnetic field produced by the coil itself is called: <u>*proper flux*</u>. The proper flux can be calculated by the equation:

$$\phi = Li$$

Where L is called the inductance of the coil. L is a constant which is a characteristic of the coil itself. L is measured in SI in henries (H); this is named after the American physicist Joseph Henry. For a coil L is calculated by:

$$L = \frac{\mu_0 N^2 S}{l}$$

S is the area of one of the turns of the coil.

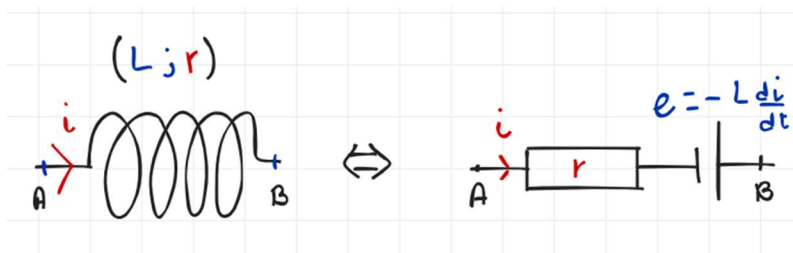
Now, when the current i through the coil varies, the magnetic field produced by the coil and therefore the proper flux varies. The coil reaction is to produce an e.m.f, that we call this time an auto or a self-induced e.m.f (since the source of the field and the induced circuit are same). The self-induced e.m.f can be found, according to Faraday's Law by:

$$e = -\frac{d\phi}{dt} = -\frac{d(Li)}{dt} = -L\frac{di}{dt}$$

Self-induction video 1: https://youtu.be/DRXkAOgR51s

Self-induction video 2: https://youtu.be/Et0aaDDQpyI

As result, a coil traversed by a variable current behaves like a generator of electromotive force e (the self-induced e.m.f) and an internal resistance r (the resistance of the wire the coil is made of).



By Ohm's Law applied to a coil oriented from A to B:

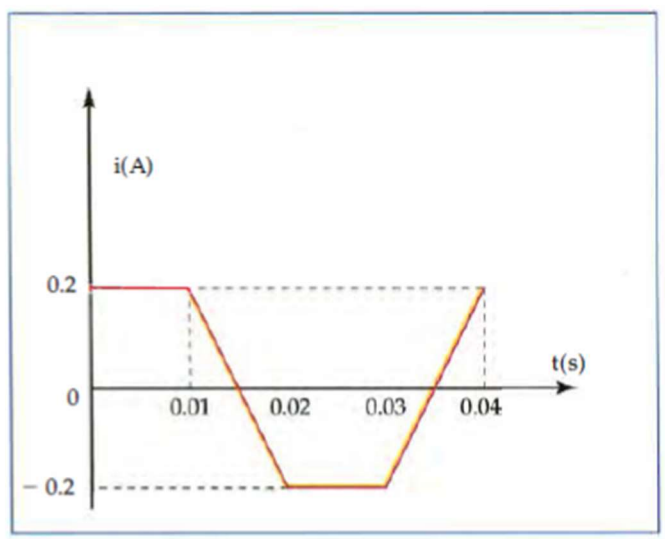
$$u_{AB} = ri - e \Rightarrow u_{AB} = ri + L\frac{di}{dt}$$

The magnetic energy stored in the coil at the moment where the current through it is i:

$$W_m = \frac{1}{2}Li^2$$

Application:

A part (A, B) of a circuit is composed of a coil (without a soft iron core), of inductance L = 5.0 mH, and of resistance r = 2 Ω. The coil is oriented from A to B.



1) Calculate the value of the self-flux through the coil when this is traversed by a current I = 0.2 A.

2) The coil is traversed by a current which varies with time as indicated by the figure for $0 \leq t \leq 0.04\ s$.

a) For what intervals of time does the proper flux through the coil vary?

b) Deduce that the coil is the seat of self-induced e.m.f in the specified intervals of time. Calculate this e.m.f. in each of these intervals.

c) Give the expression for the potential difference u_{AB} across the coil. Determine its value in each interval.